

cse15l-lab-reports

Lab Report 4: Doing it All from the Command Line

Tasks:

1. Log into `ieng6`

```
$ ssh cs15lsp23if@ieng6.ucsd.edu
Last login: Wed Mar 15 20:18:26 2023 from 100.83.78.253
===== NOTICE =====
Authorized use of this system is limited to password-authenticated
usernames which are issued to individuals and are for the sole use of
the person to whom they are issued.
  https://support.ucsd.edu/its

You may also report issues, suggestions, or feedback by e-mailing root on any system:
mail -s "Your subject here" root
Type your message - Ctrl+D to send

*** Access our Linux ssh terminals or remote desktops via a web browser at: ***
https://linuxcloud.ucsd.edu

All accounts must be enrolled in Duo for access. No VPN required.

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quota: Cannot resolve mountpoint path /home/linux/staff/.snapshot/hourly.2023-05-08_1201: Stale file handle
quota: Cannot resolve mountpoint path /home/linux/staff/.snapshot/hourly.2023-05-07_0801: Stale file handle
quota: Cannot resolve mountpoint path /home/linux/staff/.snapshot/daily.2023-05-05_0010: Stale file handle
Hello cs15lsp23if, you are currently logged into ieng6-203.ucsd.edu

You are using 0% CPU on this system

Cluster Status
Hostname    Time    #Users  Load  Averages
ieng6-201   01:35:01  0  0.91, 0.93, 1.14
ieng6-202   01:35:01  1  1.04, 1.12, 1.13
ieng6-203   01:35:02  6  0.04, 0.11, 0.20

Sun May 21, 2023  1:38am - Prepping cs15lsp23
```

- Keys pressed: `<up><enter>`
- Typed Command: `ssh cs15lsp23if@ieng6.ucsd.edu`

- Explanation: To log into my `ieng6`, I typed in the command `ssh cs15lsp23if@ieng6.ucsd.edu` to remotely connect to the server. Since I had setup an SSH key to log into this account, I didn't need to type in my password this time and was able to login immediately.

2. Clone fork of repository from Github account

```
[cs15lsp23if@ieng6-203]:~:211$ git clone git@github.com:jxnlee/lab7.git
Cloning into 'lab7'...
Warning: Permanently added the RSA host key for IP address '140.82.112.4' to the list of known hosts.
remote: Enumerating objects: 55, done.
remote: Counting objects: 100% (20/20), done.
remote: Compressing objects: 100% (13/13), done.
remote: Total 55 (delta 8), reused 17 (delta 7), pack-reused 35
Receiving objects: 100% (55/55), 376.53 KiB | 1.77 MiB/s, done.
Resolving deltas: 100% (20/20), done.
```

- Keys pressed: `<Ctrl> r` + `'git cl'` + `<enter>`
- Typed Command: `git clone git@github.com:jxnlee/lab7.git`
- Explanation: Since I had previously performed these tasks before, to clone the repository I only needed to type `<Ctrl> r` to search through my history of commands and type the first few letters of the command in order to fully type the command `git clone git@github.com:jxnlee/lab7.git` to clone the repository.

3. Run failing tests

```
[cs15lsp23if@ieng6-203]:~:212$ cd lab7
[cs15lsp23if@ieng6-203]:lab7:213$ bash test.sh
JUnit version 4.13.2
..E
Time: 0.544
There was 1 failure:
1) testMerge2(ListExamplesTests)
org.junit.runners.model.TestTimedOutException: test timed out after 500 milliseconds
    at ListExamples.merge(ListExamples.java:44)
    at ListExamplesTests.testMerge2(ListExamplesTests.java:19)

FAILURES!!!
Tests run: 2, Failures: 1

[cs15lsp23if@ieng6-203]:lab7:214$
```

- Keys pressed: `'cd l'` + `<tab><enter>` + `<Ctrl> r` + `'ba'` + `<enter>`
- Typed Command: `cd lab7` + `bash test.sh`
- Explanation: Before running the tests, I needed to enter the directory for `lab7`, since I knew this directory was now in my current directory and was the only one that started with the letter 'l', I only needed to type `cd` + `'l'` to change the directory to `lab7`. To run the tests, I simply needed to search through my history of commands with `<Ctrl> r` with the first few letters for the

command `bash test.sh` to run the tests. As indicated in the screenshot, there is a bug in the `ListExamples.merge` method which produces an error in the `testMerge2` test.

4. Edit code to fix failing test

```
[cs15lsp23if@ieng6-203]:lab7:219$ vim ListExamples.java
```

- Keys pressed: `<Ctrl> r` + `'vi'` + `<enter>`
- Typed Command: `vim ListExamples.java`
- Explanation: To edit the code, I first needed to enter `vim` for `ListExamples.java`. Since I had performed this task earlier, I searched through my history of commands with `<Ctrl> r` along with the first few letters of the `vim ListExamples.java` command to access the code.

```

}
while(index1 < list1.size()) {
    result.add(list1.get(index1));
    index1 += 1;
}
while(index2 < list2.size()) {
    result.add(list2.get(index2));
    // change index1 below to index2 to fix test
    index1 += 1;
}
return result;
}

```

- Keys pressed: `:/cha` + `<enter><down><right><right>` + `'x'` + `'i'` + `'2'` + `<esc>` + `':wq'`
- Typed Command: (commands to navigate through vim)
- Explanation: To edit the bug in the code, I first navigated to the error by searching for the comment indicating the location of the error which states "change index 1 below to index2 to fix test" with the bug located in the line below. First I typed `/` to use the search command in vim to locate the word "change" and navigated to its location with the first few characters. Then I used the arrow keys to navigate specifically to where the bug is located, pressed the `x` command to delete the character '1' in 'index1', entered 'insert mode' with the vim command `i` and then replaced the deleted character with '2'. To save these changes, I exited insert mode by pressing `<esc>` and typed the `:wq` command, which saves the changes and quits vim.

5. Run successful tests

```
[cs15lsp23if@ieng6-203]:lab7:220$ bash test.sh
JUnit version 4.13.2
..
Time: 0.014

OK (2 tests)
```

- Keys pressed: `<Ctrl> r` + `'ba'` + `<enter>`
- Typed Command: `bash test.sh`
- Explanation: To verify that the change is successful, I run the tests the same way I demonstrated the failed tests, by searching for the `bash test.sh` command with `<Ctrl> r`. As shown in the screenshot, all tests successfully passed.

6. Commit & pushing resulting changes to Github account

```
[cs15lsp23if@ieng6-203]:lab7:221$ git add ListExamples.java
[cs15lsp23if@ieng6-203]:lab7:222$ git commit -m "fixed ListExamples"
[main 499454d] fixed ListExamples
Committer: Jeffrey S Lee <cs15lsp23if@ieng6-203.ucsd.edu>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 1 insertion(+), 1 deletion(-)
[cs15lsp23if@ieng6-203]:lab7:223$ git push origin main
Warning: Permanently added the RSA host key for IP address '140.82.113.4' to the list of known hosts.
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 309 bytes | 309.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:jxnlee/lab7.git
503ed81..499454d  main -> main
```

- Keys pressed: `<Ctrl> r` + `'git a'` + `<enter>` + `<Ctrl> r` + `'git c'` + `<enter>` + `<Ctrl> r` + `'git p'` + `<enter>`
- Typed Command: `git add ListExamples.java` + `git commit -m "Fixed ListExamples"` + `git push origin main`
- Explanation: Since I had used these git commands before in the tasks earlier, I only needed to search for each of the commands with `<Ctrl> r` and the first few letters of each command. So `<Ctrl> r` + `'git a'` typed the `git add ListExamples.java` command, `<Ctrl> r` + `'git c'` typed the `git commit -m "Fixed ListExamples"` command, and `<Ctrl> r` + `'git p'` typed the `git push`

`origin main` . The first one added the file to the changes that needed to be committed and pushed, the second command commits the results, and the last command pushes the results to the repository to save the changes to the main branch.

Note: Since I had performed the tasks in the account earlier, I am able to quickly put in the commands by searching through my history, otherwise I would alternatively type in the commands as I would normally without the history